Online Resource 1

Supplementary material for the article "AXIOM: A Shared Graph-Attention Controller for Scalable Multi-Agent 5G Slice Admission"

Wireless Personal Communications

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S1. Overview

This supplement gives the full results of the verification runs that the main paper summarises in its section on reproducibility and fresh-seed verification. Every pass criterion was fixed in a pre-registration committed before the run it judged, and every check is reported whether it held or failed. The baselines are a single-agent deep Q-network (DQN-SA), independent per-gNB deep Q-networks (DQN-I), and multi-agent proximal policy optimization (MAPPO). AXIOM's encoder is a graph attention network throughout, and the MLP variant replaces it with a per-node multi-layer perceptron (MLP), as Section S12 describes.

S2. Reruns on the original seeds

We reran eleven reported cells on their original seeds in fresh processes. AXIOM in both environments and at nineteen gNBs, the MLP variant, MAPPO, DQN-I, the tuned rules, federated training, and the dropout cell all reproduced their stored per-seed results exactly. The rule we rely on as the Env A oracle also matched the stored oracle exactly on all 30 seeds.

DQN-SA is the one exception to exact reproduction. Its training code builds the network before it sets the seed, so the first seed in each batch of runs starts from an unseeded initialisation, while later seeds in the batch reproduce exactly. Five reruns of one such seed earned 7.45 to 9.67 reward per step, against 8.74 stored. Dropping the affected seeds leaves every comparison with this baseline in the same direction. The one change is in Env A at three gNBs, where AXIOM's lead moves from +0.311 to +0.241 per step but now passes the Holm correction, at $p = 0.039$ against 0.066.

S3. First fresh-seed run

We retrained every arm on fresh seeds from 3000 onwards, never used before, with 20 seeds at three gNBs, 15 in Env A at nineteen gNBs, and 10 in Env B at seven gNBs. All 26 predictions stated in advance for these fresh seeds held. In Env A at three gNBs AXIOM reached 98.2% of the oracle reward and beat DQN-I and MAPPO at Holm-adjusted $p \leq 0.002$. At nineteen gNBs it kept 95.6% priority-slice compliance against 47.8% to 65.5% for the baselines, winning all six tests at Holm-adjusted $p \leq 0.001$.

In Env B at three gNBs AXIOM beat the static and accept-all rules in all 20 seeds and led every learned baseline at Holm-adjusted $p \leq 0.0002$. The tuned local rule again beat AXIOM, earning 9.175 against 8.782 per step, and at seven gNBs it beat every learned policy at $p \leq 0.004$. Two fresh-seed results are less favourable to AXIOM than the main campaigns. In Env A at

three gNBs AXIOM won 11 of 20 seeds against DQN-SA but trailed it by 0.021 per step on average, at $p = 0.07$, so the pooled 40-seed lead over that baseline does not carry over to new seeds.

In Env B at seven gNBs the per-node MLP encoder beat graph attention by 1.38 reward per step, in 9 of 10 seeds at $p = 0.006$, which strengthens the finding that attention adds nothing.

S4. Second fresh-seed run

The second fresh-seed run, also specified in advance, covered Env A at seven gNBs under the ring and fully-connected layouts, at nineteen gNBs under the hex layout, and federated training, each on ten seeds. At nineteen gNBs AXIOM kept 96.5% priority-slice compliance against 47.5% to 65.2% for the baselines, winning all six tests at Holm-adjusted $p \leq 0.012$. Federation again showed no significant cost, at +0.077 with 95% CI [−1.03, 1.27].

Seven gNBs gave the weakest result among all the fresh-seed checks. AXIOM kept 96.7% priority-slice compliance and led every baseline under both layouts, and all six ring comparisons held at Holm-adjusted $p \leq 0.032$. Under the fully-connected layout, however, two weak AXIOM seeds at 57.2% and 76.9% compliance left its leads of 12.5 to 25.3 points short of significance after Holm correction over the twelve tests. DQN-I and DQN-SA also reached 80.9% and 80.6% compliance, above the 80% bound we had set, so three of the eleven checks in this run failed.

S5. Third fresh-seed run

The third fresh-seed run, again specified in advance, covered every remaining claim, and 43 of its 48 checks held. At seven gNBs under the hex layout and at nineteen gNBs under the fully-connected layout, AXIOM kept 99.0% and 95.7% priority-slice compliance and beat every baseline in all six tests at Holm-adjusted $p \leq 0.012$. A separate twenty-seed rerun of the seven-gNB fully-connected layout, reported alongside the failed ten-seed check rather than in its place, also held all six tests at Holm-adjusted $p \leq 0.007$.

Three of the five failed checks concern the baselines rather than AXIOM. On fresh seeds at seven gNBs DQN-I and DQN-SA kept 76.2% to 83.3% priority-slice compliance, above the 80% bound in three cells, so the baseline range of 56% to 76% holds only for the main campaign. The other two failures come from the load sensitivity study. At 102% of the nominal load two weak AXIOM seeds left it at 96.8% of the oracle reward, a gap that is not significant at $p = 0.25$, and below DQN-SA's mean, at 12.551 against 12.910. Under 107% load and the changed weights AXIOM stayed within 0.6% of the oracle.

In Env B at seven gNBs under the fully-connected and hex layouts, the tuned local rule again beat every learned policy at $p \leq 0.004$, AXIOM was not significantly better than the static rule, and the MLP encoder again earned more than graph attention, though not significantly. Under disruption every dropout and churn cost was significant at Holm-adjusted $p \leq 0.041$, no gap between AXIOM and a baseline survived correction, and AXIOM kept a spike block precision of 0.97. Per-step differential-privacy noise lowered reward by 0.65 to 1.31 per step with no drop surviving correction, and every client took the same 17,969 noisy steps, so the quoted ε values apply unchanged. In the Env B sensitivity study the original AXIOM again earned less than the current design in all ten seeds, by 2.19 per step, while the higher learning rate and the wider encoder made no significant difference.

Finally, we reran DQN-SA with its initialisation seeded, through a wrapper that leaves the framework untouched, and in a two-seed test it then reproduced exactly across processes. AXIOM's lead over it in Env A at three gNBs becomes +0.254 per step at Holm-adjusted $p = 0.018$, and +0.948 on the replication seeds. In Env B the lead stays at +1.320, and at nineteen gNBs the seeded baseline collapses in 28 of 30 seeds and trails AXIOM by 32.3 points of priority-slice compliance. Across these samples AXIOM's lead over DQN-SA in Env A at

three gNBs is therefore small and unstable, significant in some samples and absent in the fresh ones.

S6. Fourth fresh-seed run

The last fresh-seed run, specified in advance, enlarged the two unstable samples, and four of its five checks held. On 40 new seeds in Env A at three gNBs AXIOM reached 98.1% of the oracle, tying it in 27 seeds, and beat DQN-I and MAPPO at Holm-adjusted $p \leq 0.0001$. Its lead over the seeded DQN-SA was only +0.159 per step, at $p = 0.090$. On 20 new seeds at 102% load AXIOM came within 0.5% of the oracle, but the seeded DQN-SA again earned slightly more, 12.978 against 12.945 per step. On fresh seeds AXIOM therefore does not lead DQN-SA under this condition, contrary to the main sensitivity sample.

S7. Independent recomputation

A script written independently of the analysis pipeline, with its own log reader and Holm correction, recomputed from the raw logs the numbers in every results table and the key results sentences, including those of this supplement. All of its checks pass, as the main paper reports, after we corrected the three rounding errors it found, and every quoted confidence interval matches a fresh bootstrap.

S8. Summary

Reruns on fresh seeds confirmed the main findings of the paper. The exceptions are a small and unstable lead over DQN-SA in Env A at three gNBs and baseline compliance of up to 83% at seven gNBs. In one ten-seed load sample two weak AXIOM seeds also left it 3.2% short of the oracle, a gap that a twenty-seed rerun did not reproduce.

S9. Live testbed and deployment check

The concrete rig behind every live measurement is the single-host, software-radio, O-RAN stack of Fig. 1, which runs without a near-real-time RAN Intelligent Controller (RIC). We omit the RIC because the ORANSlice fork applies slicing control directly through a lightweight UDP E2 agent, which exercises the real per-slice PRB-ratio primitive without the overhead of a separate RIC on a single commodity host.

This section reports what the real OpenAirInterface (OAI) testbed of Fig. 1 can and cannot establish about the offline-trained policy. AXIOM itself is not deployed on the live testbed. Its multi-gNB policy needs at least two live gNBs, and our two-gNB attempt registered both gNBs with the core but did not sustain traffic, so a live AXIOM run is left to future work. This section therefore validates the deployment pipeline that AXIOM would use, namely the E2 control path, the PRB-ratio actuation, and real-time inference of an offline-trained checkpoint. For that we run a single-agent DQN checkpoint, seed 900, against a real gNB, real User Equipments (UEs), and real per-slice User Datagram Protocol (UDP) traffic, at one UE per slice and again at two UEs per slice. This checkpoint was trained offline for this rig and is distinct from the three-gNB DQN-SA baseline.

A capacity probe at a deliberately wide-open ceiling showed all three slices serving essentially 100% of offered demand with 0.000% loss at both loads. The frozen checkpoint's decision logic runs correctly end to end, through the real E2 control loop, with real accept and reject decisions, and its signature low-priority-only blocking of mMTC reappears at both loads, which shows that the deployment pipeline works end to end. Because the rig never actually runs short of

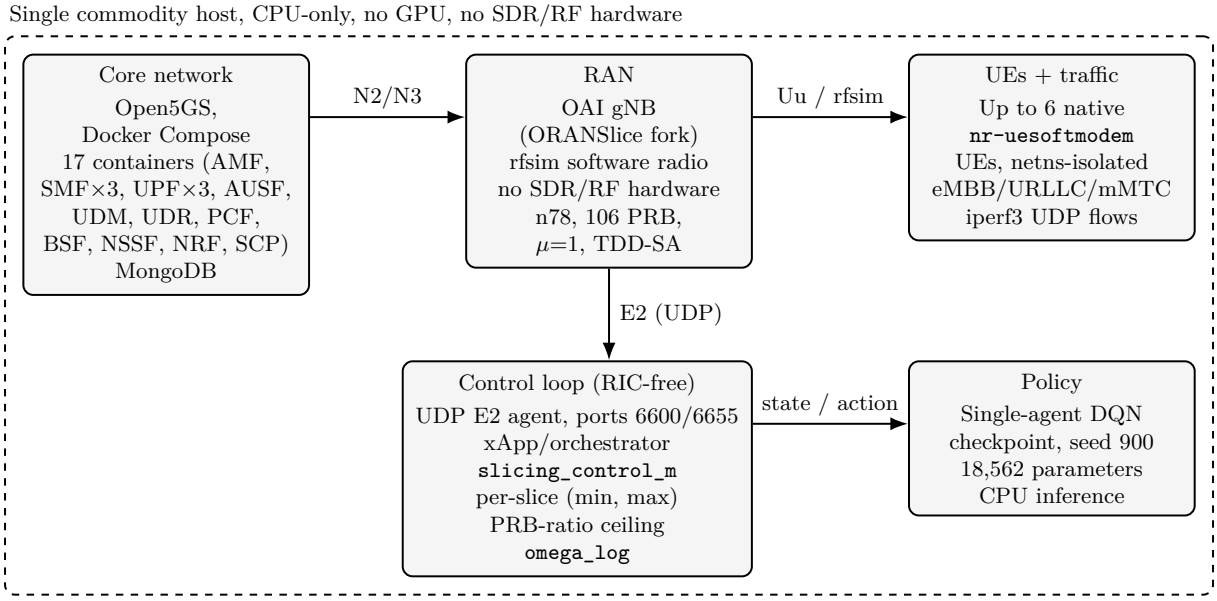


Figure 1: Real testbed stack, a single-host, software-radio rfsim, RIC-free O-RAN deployment distinguished from software-defined-radio hardware-in-the-loop testbeds. Every block and label is a measured or configured value from this rig

capacity, the same pattern is equally consistent with the policy's trained priority bias, so we claim admission-control behaviour under contention only in the offline shared-budget environment, not live. Two integrity issues surfaced and were fixed during these checks. A scheduler-configuration error had turned the per-slice PRB-ratio ranges into raw grants below the gNB scheduler's hard five-PRB floor, which we fixed at the root, after which direct instrumentation confirmed zero below-floor grants across more than 100,000 readings. A separate unit mismatch in the live SLA-margin diagnostic was corrected by converting backlog into a queueing-delay estimate, using Little's law, against each slice's own latency budget.

S10. Additional setup details

Fig. 2 gives an overview of the research, and Table 1 lists the simulation and training parameters shared by both environments. Algorithm 1 of the main paper states AXIOM's training loop.

All learned arms share the discount factor $\gamma = 0.95$ and the training budget. The value-based arms also share the Adam optimiser, an epsilon schedule decaying from 1.0 to 0.05 at a per-episode factor of 0.985, and target synchronisation every 10 episodes. AXIOM and the DQN baselines train at learning rate 10^{-3}. The GAT encoder uses hidden and output dimension 16, feeding per-slice Q-heads of hidden dimension 32. A sensitivity study repeats the learned arms and the oracle rule in Env A on 10 seeds under three changed conditions. Two change the load, to 102% and 107% of the nominal shares, and the third lowers eMBB's priority weight from 3.5 to 2.5 and raises mMTC's from 0.3 to 0.6. A second study repeats AXIOM in Env B on 10 seeds with learning rates of 3×10^{-4} and 3×10^{-3} and with an encoder twice as wide.

The federated campaign runs in Env A on seeds 900 to 909, with a five-level noise sweep $\sigma \in \{0, 0.5, 1, 2, 4\}$ and a replication on seeds 1000 to 1009. Each federated round runs 50 local steps under FedAvg with a gradient-clip norm of 1.0.

A validation screen checks every trained checkpoint of every learned arm before evaluation. Each checkpoint plays five greedy episodes on its own training stream, never on the held-out stream, and fails the screen if it blocks no request or if fewer than 90% of its blocks target mMTC. We report every seed whether or not it passes, and separately report a two-seed deployment

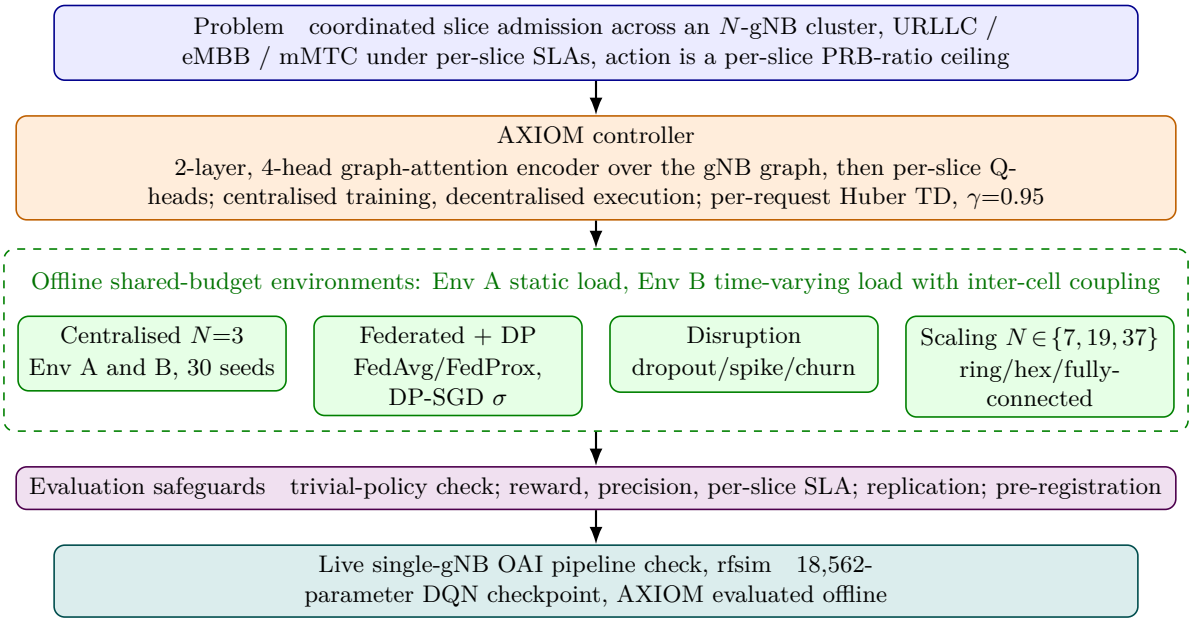


Figure 2: Overview of the research. An AXIOM controller is evaluated across four offline campaigns in two shared-budget environments, every result checked by evaluation safeguards, with the deployment pipeline checked on a live single-gNB testbed

Table 1: Simulation and training parameters, shared by both environments unless stated

Parameter	Value
PRB budget per gNB, B	100
Nominal share, floor, cap, in PRB ratio	URLLC 30, 5, 33; eMBB 60, 10, 66; mMTC 10, 5, 11
Priority weight ω_k	URLLC 12.0, eMBB 3.5, mMTC 0.3
Per-accept reward R_k	1.0 for every slice
Violation penalty λ_k	URLLC 10.0, eMBB 5.0, mMTC 2.5
Latency and loss budgets	URLLC 15 ms, 0.8%; eMBB 45 ms, 2.0%; mMTC 65 ms, 3.5%
Congestion coefficient κ, load-balance weight β	1.5, 1.0
Episode	60 steps of 5 s; queues emptied at each episode start
Requests per step	N, each with a uniformly random slice and gNB; at most $4N$ pending
Offered demand per slice	105% of nominal share, per-step noise $0.005B$, per-gNB multiplier in $[0.9, 1.0]$
Env B load and coupling	$0.95\,(1 + 0.3\sin(2\pi(s + \psi_i)/60))$, $B_i(t) = B\,(1 - 0.15\,\bar{u}_{\mathcal{N}(i)}(t-1))$
Admission gate step	5 PRB-ratio points per accept or reject
Training and evaluation	300 and 50 episodes per seed, held-out evaluation stream
Value-based learners	Adam, learning rate 10^{-3}, $\gamma = 0.95$, ϵ from 1.0 to 0.05 at $\times 0.985$ per episode, replay 10,000, minibatch 16, warm-up 32, target sync every 10 episodes
AXIOM network	Two-layer, four-head GAT, hidden and output 16, raw-feature skip, per-slice Q-heads of hidden 32, Huber loss, gradient clip 1.0
DQN baselines	Two hidden layers of 128, squared-error loss
MAPPO	Actor and critic learning rate 10^{-3}, 10 epochs, minibatch 64, clip 0.2, entropy 0.01, generalised advantage estimation $\lambda = 0.95$

protocol that trains a pair of seeds and deploys the first one to pass. The same screen applies to every learned arm, so it gives AXIOM no advantage the baselines lack. In Env B the reward-optimal policy sheds eMBB as well as mMTC, so the 90% criterion would fail good policies there. A pre-registration amendment, recorded before any result was read, therefore makes the no-blocking criterion the primary screen in Env B.

The simulator draws a fixed load multiplier in $[0.9, 1.0]$ for every gNB, so the cluster is heterogeneous, and every episode starts with empty queues. A gNB observes the inter-cell coupling of Env B only through its own queues. The two Env B constants were calibrated on pilot seeds with rules only, before any reported seed was run, and the main pre-registration fixed them together with AXIOM's learning rate and skip connection, all chosen on pilot seeds 2000 to 2009.

AXIOM's encoder is a graph-attention network over the gNB graph. At every control step each gNB forms its own feature row $\mathbf{x}_i$ of PRB-used ratio, congestion level, and normalised queue length. These rows stack into $\mathbf{X} \in \mathbb{R}^{N \times d}$, and a two-layer, H=4-head network produces one embedding per gNB. Each head h decomposes additive attention rather than scoring a concatenation. It uses a learned source vector $\mathbf{a}_s^h$ and destination vector $\mathbf{a}_d^h$, applied to the shared linear projection $\mathbf{W}^h$. Here LeakyReLU is the leaky rectifier of slope 0.2, and $\mathcal{N}_i$ is the set of neighbours of node i under the adjacency mask,

$$\varepsilon_{ij}^h = \mathrm{LeakyReLU}\big(\mathbf{a}_s^{h\top}\mathbf{W}^h\mathbf{x}_i + \mathbf{a}_d^{h\top}\mathbf{W}^h\mathbf{x}_j\big), \qquad \alpha_{ij}^h = \frac{\exp \varepsilon_{ij}^h}{\sum_{l\in\mathcal{N}_i} \exp \varepsilon_{il}^h}. \tag{1}$$

Here ε_{ij}^h is the unnormalised attention score from node i to neighbour j in head h, and α_{ij}^h is its normalisation over i's neighbours, with l ranging over $\mathcal{N}_i$. Hidden layers concatenate the H heads under an exponential linear unit (ELU) nonlinearity, while the final layer averages them. A per-layer normalisation operator $\mathrm{LN}(\cdot)$ then rescales each node's embedding to zero mean and unit variance, and produces the final per-gNB embedding $\mathbf{z}_i$,

$$\mathbf{z}_i = \mathrm{LN}\Big(\tfrac{1}{H}\textstyle\sum_{h=1}^{H}\sum_{j\in\mathcal{N}_i}\alpha_{ij}^h\,\mathbf{W}^h\mathbf{x}_j\Big). \tag{2}$$

Centralisation comes from this encoder, whose parameters every agent's loss updates jointly, so it sees the full cluster state during training. Decentralised execution comes next, since at action-selection time each agent consumes only its own embedding row $\mathbf{z}_i$ to decide its own pending requests.

An earlier AXIOM variant passed each gNB's summed per-request values through a QMIX-style monotonic mixer. Because the number of pending requests changes from step to step, that mixed target changed scale with the request count, and the mean training loss diverged, rising from about 76 over the first fifth of training steps to about 409 over the last fifth. Treating each request as its own sample removed that dependence, and the Huber loss keeps the deeper encoder stable under long bootstraps.

In the federated variant each client's encoder still reads the full joint node-feature snapshot, and FedProx, which adds a proximal term $\frac{\mu}{2}\|\theta_i - \bar{\theta}\|^2$, gave no consistent improvement at $\mu \in \{0.01, 0.1, 1.0\}$ on three seeds. Differential privacy bounds how much any one client's training data can change what the shared model reveals. Each client applies a differentially private stochastic gradient descent (DP-SGD) step, clipping its minibatch gradient to norm C_{dp} and, when privacy is enabled, adding Gaussian noise of standard deviation σC_{dp}. Because the clip covers the whole minibatch, changing a client's data in one step moves the clipped gradient by at most $2C_{\mathrm{dp}}$, so the guarantee covers the client's step batch. Zero-concentrated differential privacy (zCDP) composes the Gaussian mechanism over the m noisy steps a client takes, with ε and δ the usual privacy-loss and failure-probability parameters,

$$\rho = \frac{2m}{\sigma^2}, \qquad \varepsilon(\delta) = \rho + 2\sqrt{\rho \ln(1/\delta)}. \tag{3}$$

We credit no amplification by subsampling, because minibatches come from a replay buffer that reuses records, so the bound is conservative. The clip applies whether or not noise is added, so runs with σ=0 and σ>0 differ only in the noise, which protects each client's admission decisions and rewards but not the congestion features that every encoder reads by design.

An AI assistant, Anthropic's Claude, was used to help write the analysis and plotting scripts and to edit the manuscript text. The authors made all research decisions and take full responsibility for the content of this paper.

S11. Evaluation safeguard details

The main paper uses three evaluation safeguards, namely a trivial-policy check, correctness-aware metrics, and replication on disjoint seeds. The main paper defines the metrics in full, and this section gives the details of the other two.

The trivial-policy check evaluates the simplest rules the environment admits before trusting any learned policy, namely accepting every request and shedding one slice outright. In the first version of our simulator, every slice was served independently against its own ceiling at about a quarter of the gNB's budget. There, accepting every request earned 14.45 reward per step and kept URLLC and eMBB at 100% SLA compliance, more than any learned policy, so admission control had nothing to do and policies that appeared to fail by never blocking were in fact optimal. The shared-budget simulator of the main paper removes that loophole. Over the same 30 held-out seeds, accepting every request now earns 9.89 reward per step and lets eMBB compliance fall to 56.8%, whereas shedding every mMTC request earns 12.84 and keeps URLLC and eMBB at 100%. Every result in this paper comes from the shared-budget simulator.

The replication safeguard is a reproduction-versus-replication discipline, in which we recompute every headline comparison on an independent, disjoint seed sample and report both. Reproduction re-runs the same seeds, whereas replication draws fresh ones, so only replication can catch a result that is an artifact of one seed sample. It caught one such artifact in this work. A weaker replication led us to inspect step-level traces, which showed that queue backlog carried over between episodes. With only about half a PRB of headroom once mMTC is shed, a backlog built by one early transient never drained, so a single bad episode spoiled every later one for every policy, the oracle rule included. Every result in this paper comes from the corrected simulator, which empties all queues at the start of each episode.

S12. Additional results

In the decision map of the main paper's nineteen-gNB hex cluster, which has the same layout as its thirty-seven-gNB map, AXIOM keeps the URLLC and eMBB gates at 1.00 and 0.97, while closing the mMTC gate to 0.08. DQN-SA leaves every gate open at 0.99, DQN-I closes the mMTC gate only to 0.92, and MAPPO partly closes the eMBB gate, to 0.82, while leaving mMTC open at 0.69.

The MLP variant replaces AXIOM's graph-attention encoder with a per-node MLP that reads only a gNB's own features, and it was run in four cells. Table 2 sets it beside AXIOM at three gNBs. In Env A it ties AXIOM, which earns 0.058 more reward per step at three gNBs and 0.31 more at nineteen gNBs on the ring layout. In Env B it earns slightly more, by 0.116 per step at three gNBs, in 17 of 30 seeds at $p = 0.45$, and by 1.12 and 0.95 at seven gNBs, where neither difference survives Holm correction. On ten fresh seeds at seven gNBs its lead of 1.38 per step was significant, as Section S3 reports. The variant uses 3,190 parameters at every cluster size and takes 0.77, 1.69, 4.46, and 9.44 ms per control step at 3, 7, 19, and 37 gNBs, with a median training time of 503 s per seed in Env B at three gNBs. It was not run in any later campaign, so every result at thirty-seven gNBs, under overload, and for throughput and delay uses the attention encoder.

Table 2: AXIOM and its MLP variant at three gNBs, 30 seeds per arm, 95% bootstrap CIs. Columns as in the main paper's three-gNB tables

Env A	Reward/step	Precision	Collapsed	URLLC	eMBB
AXIOM	12.698 [12.49, 12.86]	0.992	0/30	0.998	0.979
AXIOM, MLP encoder	12.640 [12.46, 12.79]	0.986	0/30	1.000	0.976
Env B	**Reward/step**	**URLLC**	**eMBB**	**mMTC**	
AXIOM	8.832 [8.64, 9.03]	0.958	0.344	0.406	
AXIOM, MLP encoder	8.948 [8.77, 9.13]	0.974	0.337	0.452	

A disjoint replication on seeds 1000 to 1009 is stronger for AXIOM. There it earns 1.083, 0.799, and 3.806 more reward per step than DQN-SA, DQN-I, and MAPPO, at Holm-adjusted $p \leq 0.023$, while sitting only 0.018 below the oracle. Pooled over all 40 seeds, its margins over the three baselines are 0.504, 1.032, and 2.965 reward per step, at Holm-adjusted $p \leq 0.002$, and falls 0.108 short of the oracle, a small gap that is significant at $p = 0.034$. Fresh seeds in Sections S3 to S6 do not reproduce the lead over DQN-SA, however.

AXIOM's behaviour also holds in Env A when the environment changes. Under three changed conditions, namely load at 102% and 107% of the nominal shares and the changed slice weights, AXIOM stays within 0.13 of the oracle rule's reward with no significant difference. On fresh seeds the gap at 102% load was larger in a ten-seed sample but not in a twenty-seed one, as Section S5 reports. On these seeds it keeps a positive lead over both DQN baselines under every condition, but at ten seeds only two of those six leads survive correction. They are its margin over DQN-I at 107% load, 1.073 reward per step at $p = 0.006$, and over DQN-SA under the changed weights, 1.264 at $p = 0.041$.

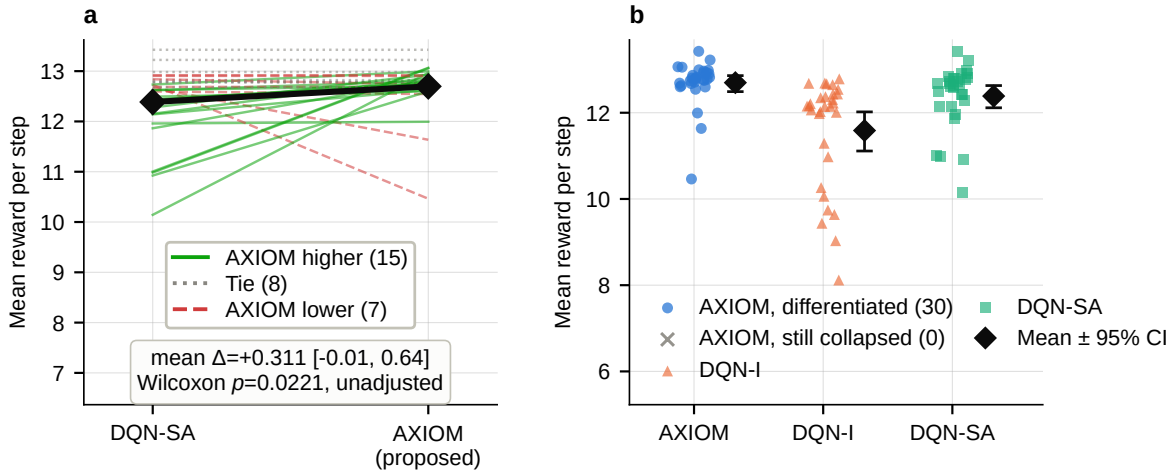


Figure 3: Env A, three gNBs, per-seed evidence. a, Paired per-seed mean reward per step of AXIOM versus DQN-SA, coloured by win, tie, or loss. b, Per-seed reward distribution for AXIOM and the two DQN baselines

The original AXIOM, at learning rate 10^{-4}, came slightly closer to the oracle in this environment, tying it in 39 of 40 seeds. The change to the current design therefore trades a little static-load accuracy for the ability to adapt that the time-varying environment rewards.

The learning rate matters for AXIOM and its baselines alike. Every learned arm now trains at 10^{-3}, and the DQN baselines lose reward when rerun at 10^{-4}, falling to 11.477 and 11.119 per step. AXIOM depends on its rate even more in Env B. There the original AXIOM, at 10^{-4}, earns only 6.496 per step on seeds 900 to 909, 2.37 below the current AXIOM in all ten seeds,

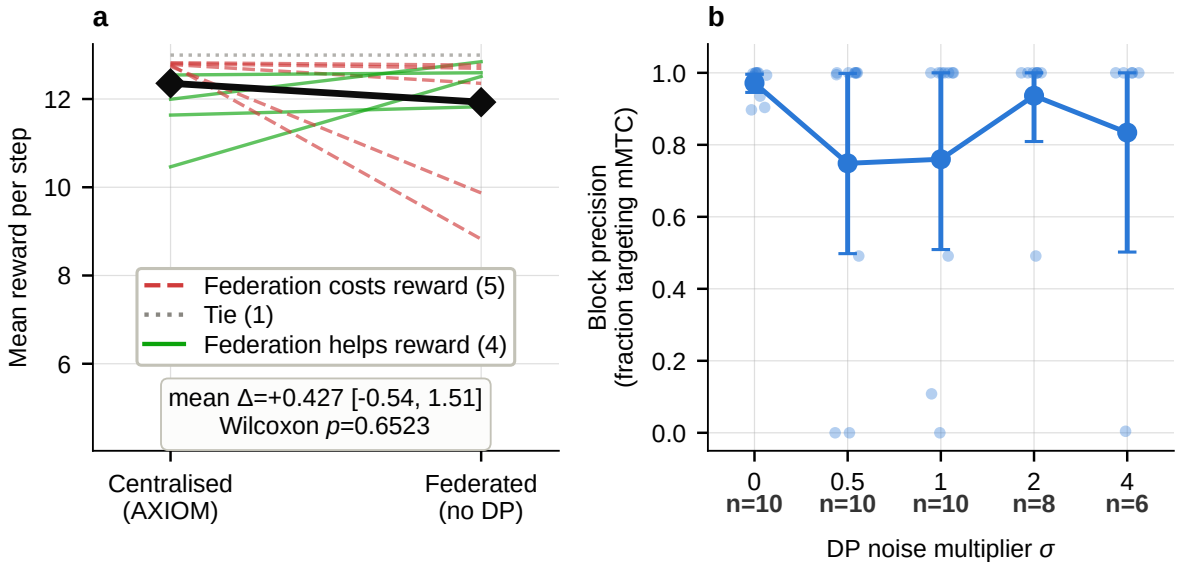


Figure 4: Federated campaign in Env A, 10 seeds. a, Paired per-seed federation cost on mean reward per step, not significant at $p = 0.65$. b, Block precision versus DP noise multiplier σ, over the seeds that block

and a rate of 3×10^{-4} reaches 7.220. A rate of 3×10^{-3} and an encoder twice as wide both stay within 0.19 of the current AXIOM, with no significant difference. At nineteen gNBs in Env A the original AXIOM kept priority-slice compliance at only 79.1%, so the faster learning rate matters most at scale.

With federation switched on and no noise, mean reward per step on the ten shared seeds is 12.356 for centralised AXIOM and 11.929 for the federated variant. That is a difference of +0.427 with 95% CI [−0.54, 1.51] and $p = 0.65$, so federation shows no significant cost on these seeds, although it is more variable, with five seeds losing and four gaining. The disjoint replication shows a larger cost of +0.968, with 95% CI [0.18, 1.96], which falls just short of significance at $p = 0.055$.

FedProx on three seeds gives no consistent picture, with μ=0.01 above FedAvg and μ of 0.1 and 1.0 below it. The DQN baselines are not federated, because averaging the independent learners' weights would turn them into a parameter-sharing learner that is no longer the ablation, and DQN-SA is one central controller with nothing to federate.

Adding DP-SGD noise to every step lowers reward by 0.77 to 1.25 per step against centralised AXIOM at every tested noise multiplier, from σ=0.5 to σ=4. It also lowers eMBB compliance to between 74.3% and 83.0%, and two and four of the ten seeds collapse at σ of 2 and 4. None of these drops survives Holm correction at ten seeds, as Fig. 4b reflects. The formal guarantee is weak at these settings. Each client takes 17,969 noisy steps, so the zCDP accounting above gives ε of about 146,000, 37,000, 9,600, and 2,600 at δ=10^{-5} for σ of 0.5, 1, 2, and 4, far from the single-digit values that carry meaning. Per-step DP-SGD therefore delivers no meaningful guarantee here. A useful guarantee would need noise on the roughly 360 per-round model updates rather than on every step, which the main paper lists as future work.

We test every policy's frozen Env A checkpoint against three mid-episode disruptions, injected at evaluation time with no retraining. A 'gNB dropout' forces one gNB to reject everything and blanks its row in the shared feature matrix for a window. A 'demand spike' briefly multiplies the arrival rate from outside the policy. An 'agent churn' swaps one gNB's decisions for a fresh, untrained copy of its own policy for a window and leaves the others intact. Churn applies to every genuinely multi-agent policy but not to the single-agent one. For dropout and

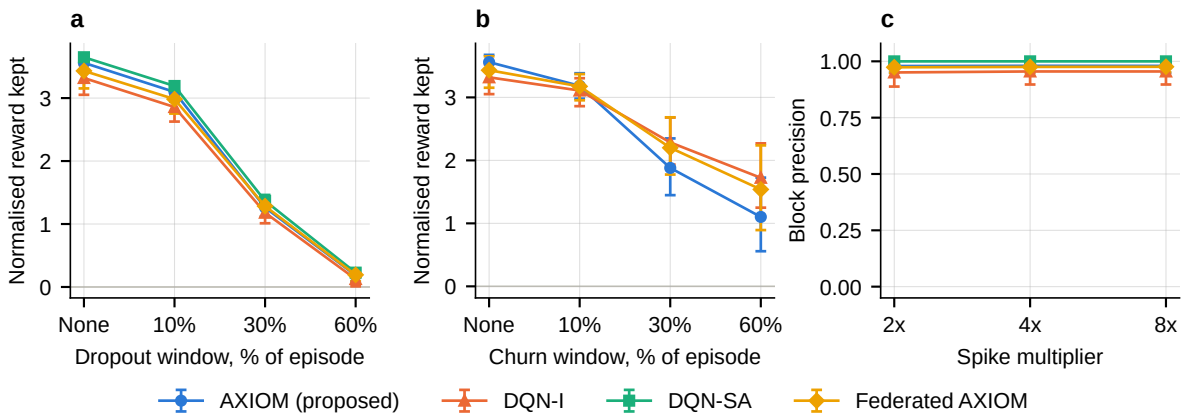


Figure 5: Disruption campaign, 10 seeds per arm, 95% CIs. a, b, Normalised reward kept under dropout and churn, with the undisrupted level at the tick marked None. c, Block precision under demand spikes

churn the severity is how long the window lasts, at 10%, 30%, or 60% of the episode. For the spike it is the arrival multiplier, at 2×, 4×, or 8×. Because raw reward grows mechanically with request volume, we report a volume-normalised reward over an episode of T steps, $\bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t / \max(n_t, 1)$, which divides each step reward r_t by its own pending-request count n_t before averaging.

Fig. 5 plots the reward each policy keeps. Undisrupted, AXIOM earns 3.56 normalised reward, DQN-SA 3.65, and DQN-I 3.32. Dropout and churn hurt every policy significantly at every severity, in 9 or 10 of 10 seeds at Holm-adjusted $p \leq 0.041$. AXIOM's cost grows from 0.46 to 3.37 across the dropout windows and from 0.38 to 2.46 across the churn windows. A dropped gNB rejects every request, including URLLC and eMBB, so its penalties dominate the episode whatever the policy does. No gap between AXIOM and any baseline survives Holm correction under any disruption. Under the two longer churn windows AXIOM keeps less reward than DQN-I, 1.88 and 1.10 against 2.29 and 1.72, though not significantly. Demand spikes leave AXIOM's decisions largely intact, with block precision at 0.98, against 1.00 for DQN-SA and about 0.95 for DQN-I. A disjoint replication on seeds 1000 to 1009 reproduces this pattern, with every dropout and churn cost significant at $p \leq 0.035$ and no significant gap between policies.

Replication first proved its worth in this campaign, on the earliest simulator. There, a ten-seed sample suggested that DQN-I was immune to churn, since it never reads another agent's features, and a disjoint replication overturned that reading with a significant cost at every severity. In the shared-budget environment the primary sample already shows that churn hurts every policy, so the corrected reading holds.

The pre-registered validation screen rarely rejects AXIOM in Env A. It fails 1 of 30 seeds at three gNBs, 1 of 45 at seven, and 5 of 90 at nineteen, including the one seed that collapses. At thirty-seven gNBs it fails 8 of AXIOM's 60 checkpoints, including all six that collapse, against 29, 27, and 27 of 30 for DQN-SA, DQN-I, and MAPPO. The screen fails 89 of DQN-SA's 90 seeds at nineteen gNBs. It also fails many MAPPO seeds, because MAPPO also blocks eMBB, so the 90% criterion flags it. In Env B no learned policy collapsed at three or seven gNBs, while at nineteen gNBs DQN-SA collapsed in 14 of 15 seeds and AXIOM in 1, as Table 7 lists.

Fig. 6 shows how the four learned policies train. MAPPO rises fastest in the first few episodes but plateaus lowest in every cell, ending at 6.4, 16.0, and 62.9 in the three panels. The value-based arms rise more slowly but further. At nineteen gNBs AXIOM keeps improving to about 85 per step over the last 20 training episodes, while the two DQN baselines settle near 71 and 74.

Fig. 7 gives the per-layout view of the Env A scaling results at nineteen and thirty-seven

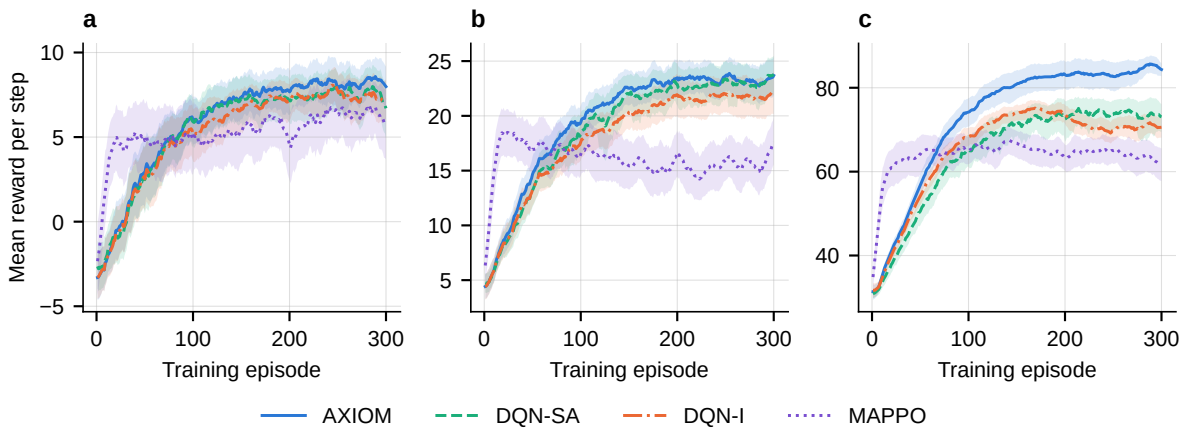


Figure 6: Training curves, mean step reward per training episode over 12 seeds with 95% bands, smoothed over 10 episodes. a, b, Env B at three and seven gNBs. c, Env A at 19 gNBs on the ring layout

gNBs that the main paper pools over layouts.

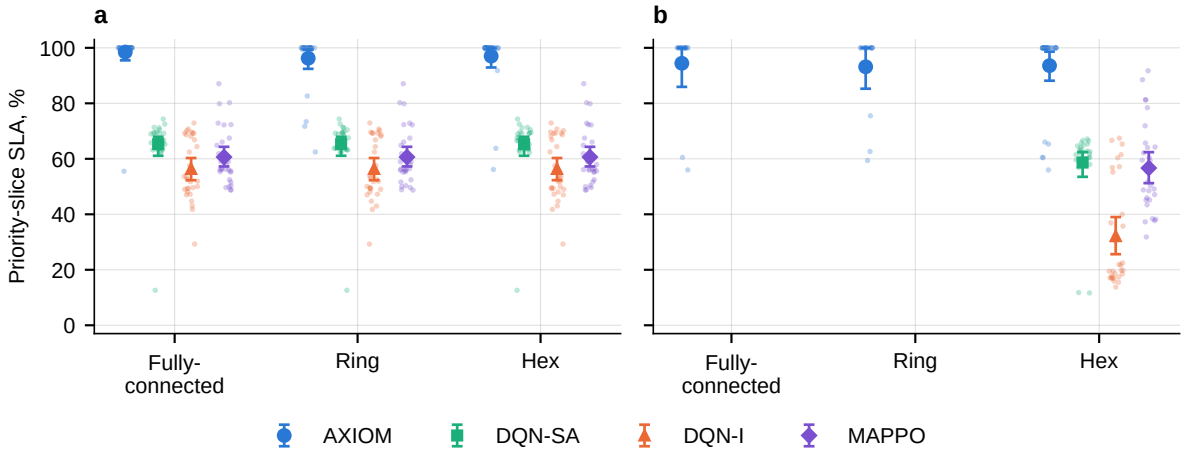


Figure 7: Env A priority-slice compliance by layout, per-seed points behind the mean and 95% CI. a, Nineteen gNBs, 30 seeds per cell. b, Thirty-seven gNBs, where the baselines never read the layout and ran once, on the hex seeds

S13. Analysis-plan commits and development notes

Each analysis plan was committed to the project's version-control history before the runs it governs, with one exception. The main campaign reuses Env A results of DQN-SA, DQN-I, and the oracle rule from an earlier rerun, M63, at three gNBs on the primary and replication seeds, in the sensitivity study, at seven gNBs, and on 15 of the 30 seeds at nineteen gNBs. The M63 plan was committed as 85f2aa9 one minute after that rerun was launched and before any of its results were read. The commits are cfc6c15 for the main pre-registration, with its screen amendment in 97fce09 and 3c8cbef, c94cf9b for the verification plan, 883219e, 19ba2b2, and 43a67f9 for its three fresh-seed addenda, and 08b6f6d for the internal-review runs. The follow-up in Section S14 was committed as a4b9856, the thirty-seven-gNB extension in Section S15 as 85e6781, its fresh-seed replication as ab76311, the runs in Section S16 as d251490, those in Section S17 as

c1f8cd2, and the measurements in Section S18 as e0184c8. Two development stages are also disclosed here, although neither changed a reported result. A protocol committed before any of its runs, M67, tried to make AXIOM's training reliable enough to beat both DQN baselines decisively in Env A at three gNBs. Selecting checkpoints on validation episodes brought AXIOM to the oracle but brought DQN-SA within 0.05 of it too, so that variant was dropped after the first of eight permitted attempts. A single test of both encoders on ten new seeds in Env B at seven gNBs then found that both led DQN-SA by 0.4 to 1.4 per step on average, none of these leads significant, so AXIOM's encoder was not changed. Separately, exploratory probes on seeds 900 to 904 suggested the load-shift effect of Section S14, and probes on pilot seeds 2000 to 2004 the hindsight bound of the main paper's Env B results, before the pre-registered runs that the paper reports.

S14. Shifted load, longer training, and the joint target on fresh seeds

No deployable policy we tested keeps eMBB compliance above 37% in Env B, because demand at the load peaks exceeds the budget by up to 23.5% and a slice counts as compliant only when every gNB meets it. On 15 seeds, even an oracle choice of the rule with the highest eMBB compliance among all 95 rules of the local family, made in hindsight on each evaluation stream, reaches only 37.7% on average. The best rule by reward in hindsight earns only 0.14% more than the tuned rule. Longer training also narrows the gap under nominal conditions. With three times the training budget AXIOM gains 0.33 reward per step at three gNBs and 2.92 at seven, on 20 and 12 seeds. The remaining gaps to the tuned rule, 0.08 and 0.96 per step, are no longer significant, at $p = 0.35$ and $p = 0.064$.

This section reports the confirmatory follow-up to the third internal review, whose plan was committed before any run as commit a4b9856. The checkpoints and rules used here were trained and tuned under nominal Env B conditions and were neither retrained nor re-tuned. Table 3 lists every shift on 25 checkpoints, seeds 905 to 929, which the exploratory probe that suggested the effect did not use. Under nominal conditions this setup reproduces the stored results exactly.

Table 3: AXIOM against the rule tuned on nominal conditions, Env B with three gNBs, 25 seeds, reward per step, wins over losses, and the unadjusted Wilcoxon p

Condition	AXIOM	Tuned rule	Difference	AXIOM wins	p
80% load	13.776	14.055	−0.279	12/13	0.44
nominal	8.846	9.261	−0.415	7/18	0.00016
105% load	4.482	4.183	+0.299	21/4	0.017
110% load	2.082	1.624	+0.458	23/2	8.8e-05
115% load	0.378	−0.052	+0.431	23/2	1e-05
120% load	−0.714	−1.068	+0.354	21/4	8.8e-05
coupling 0.30	6.457	6.766	−0.310	11/14	0.28
load swing 0.50	6.015	5.831	+0.184	18/7	0.045

The rule re-tuned on shifted training streams selected thresholds different from the nominal rule's at every shift, and against it AXIOM trails by 0.16 and 0.12 per step at 110% and 115% load, which is not significant, and by 0.62 at 120% load. The joint-target variant of AXIOM, trained from scratch beside AXIOM on fresh seeds 9000 onwards, was never significantly different from it after Holm correction over the eight tests. Its reward was 12.860 against 12.791 in Env A with three gNBs, 9.084 against 8.723 in Env B with three gNBs, 90.34 against 89.64 in Env A with nineteen gNBs, and 23.84 against 24.34 in Env B with seven gNBs. With 900 training episodes instead of 300, AXIOM earned 9.173 against 8.841 on 20 seeds in Env B with three

gNBs, where the tuned rule earns 9.249. With seven gNBs on the ring layout it earned 26.35 against 23.43 on 12 seeds, where the tuned rule earns 27.31. The training streams of these fresh seeds coincide with the evaluation streams of seeds 4000 to 4019 in the 40-seed run, which is harmless because no model is evaluated on a stream it was trained on.

S15. Thirty-seven gNBs

This section gives the setup and detail of the thirty-seven-gNB extension, whose plan was committed before any of its runs as commit 85e6781. The configuration is built from the nineteen-gNB one exactly as the seven- and nineteen-gNB configurations were built from three. Only the list of gNBs, each with 100 PRBs, the arrivals per step, now 37, and the cap on pending requests, now 148, change. Every slice, reward, and episode parameter is unchanged, and no hyperparameter was tuned for this size. Each gNB sees 1.15 pending requests per step on average, the same as at three, seven, and nineteen gNBs, and the pending cap binds in only 1 of 3,000 evaluation steps. The simulation framework defines the hex layout only for seven and nineteen cells, so a wrapper extends it to the third ring with the framework's own construction. It reproduces the seven- and nineteen-cell layouts exactly and gives the 90 edges of a three-ring hex grid. A ten-episode pilot on seed 2000 was used only to size the campaign.

AXIOM ran 30 seeds on the hex layout and 15 on each of the ring and fully-connected layouts. In Env A the simulator, the three baselines, and the static rule never read the layout, so each ran once on the 30 hex seeds and is paired with AXIOM under every layout. DQN-SA's network initialisation is seeded in these runs, and Table 4 lists every arm.

Table 4: Env A with thirty-seven gNBs. Priority-slice compliance is the mean of URLLC and eMBB compliance, and collapse counts policies that accept every request

Policy	Runs	URLLC	eMBB	Priority SLA	Reward	Collapsed
AXIOM, hex	30	97.4%	89.8%	93.6%	174.20	3
AXIOM, ring	15	93.7%	92.6%	93.2%	173.51	2
AXIOM, fully-connected	15	98.4%	90.4%	94.4%	174.44	1
DQN-SA	30	73.3%	44.1%	58.7%	144.70	26
DQN-I	30	38.6%	25.7%	32.1%	148.35	3
MAPPO	30	67.8%	45.6%	56.7%	145.25	0
Static rule	30	100.0%	100.0%	99.98%	177.58	0

AXIOM's six collapsed runs are seeds 918, 919, and 924 on the hex layout, 901 and 1001 on the ring, and 902 on the fully-connected layout. They keep 59% to 65% priority-slice compliance, while its other runs keep 96.9% to 98.1% on average under each layout. AXIOM's parameter count stays at 7,782, against 64,770 for DQN-SA and 682,058 for DQN-I. MAPPO with the configuration that the main paper's 24-configuration grid selected for Env A, trained for 900 episodes, three times every other arm's budget, ran on hex seeds 900 to 911 and 1000 to 1002 as a check the plan reported without predicting. It keeps 65.6% priority-slice compliance and 151.90 reward per step and never collapses. AXIOM beats it by 29.2 points, winning 13 of 15 seeds at $p < 0.001$, and by 23.23 reward per step, winning all 15.

The fresh-seed replication retrained every arm on seeds 6100 to 6119 on the hex layout. No earlier run used these seeds, and none of their training or evaluation streams coincides with a stream used before. Table 5 lists every arm on these seeds. AXIOM beats DQN-SA, DQN-I, and MAPPO by 37.7, 61.0, and 36.5 points of priority-slice compliance and by 35.46, 25.62, and 30.63 reward per step, winning 18 to 19 of 20 seeds in every comparison, all at Holm-adjusted $p < 0.001$. Its one collapsed seed is 6110, and pooled with the 30 hex seeds above, AXIOM keeps 94.2% priority-slice compliance over 50 seeds and collapses in 4 of them.

Table 5: Env A with thirty-seven gNBs on 20 fresh seeds, hex layout

Policy	Seeds	URLLC	eMBB	Priority SLA	Reward	Collapsed
AXIOM	20	95.4%	94.7%	95.1%	174.67	1
DQN-SA	20	71.6%	43.1%	57.4%	139.20	18
DQN-I	20	42.0%	26.1%	34.1%	149.04	2
MAPPO	20	70.1%	47.0%	58.5%	144.04	0
Static rule	20	100.0%	100.0%	99.97%	177.68	0

S16. Accept-all at scale, tuned DQN baselines, and Env B at nineteen gNBs

This section reports the runs that answer the fourth internal review, whose plan was committed before any of them as commit d251490.

Accepting every request is the floor that any admission controller should clear. In Env A it keeps 70.9%, 67.1%, and 61.9% priority-slice compliance at seven, nineteen, and thirty-seven gNBs, with reward 26.88, 78.55, and 156.22 per step. At seven gNBs only MAPPO, at 67.7%, falls below it. At nineteen and thirty-seven gNBs every baseline does, and DQN-SA's runs that do not collapse keep only 40.0% and 35.9%, DQN-I's 55.9% and 29.1%. AXIOM keeps 97.1%, 97.3%, and 93.7% at the same three sizes.

The DQN baselines were tuned on pilot seeds 2000 to 2004 over learning rates 10^{-4}, 3×10^{-4}, and 10^{-3} and minibatches of 16 and 64, keeping 300 training episodes, and each was retrained on the 30 hex seeds wherever a new setting won. Table 6 lists the selections and the results. Several DQN-SA settings tie on the pilot seeds because they collapse into accept-all there, and at nineteen gNBs the selected setting won by 0.001 reward per step. It then accepts every request in all 30 evaluation seeds and matches accept-all exactly. AXIOM beats each tuned baseline on compliance and reward in every cell, winning 26 to 30 of 30 seeds at Holm-adjusted $p < 0.001$.

Table 6: DQN baselines tuned on pilot seeds, Env A, 30 hex seeds. LR is the learning rate, batch the minibatch size, and Before the priority-slice compliance before tuning

Baseline	gNBs	LR, batch	Priority SLA	Before	Reward	Collapsed
DQN-SA	19	10^{-4}, 16	67.1%	65.3%	78.55	30
DQN-I	19	3×10^{-4}, 64	59.1%	56.4%	77.07	11
DQN-SA	37	10^{-3}, 16, unchanged	58.7%	58.7%	144.70	26
DQN-I	37	10^{-4}, 64	45.5%	32.1%	150.42	18

Table 7 lists Env B at nineteen gNBs on the hex layout, over seeds 900 to 911 and 1000 to 1002. Per-gNB compliance counts a slice as compliant at a gNB and step when that gNB meets the slice's SLA under the framework's own per-gNB test, which the cluster-level metric ORs over all gNBs. Recording it left every logged quantity unchanged, which an Env B rule evaluation at three gNBs confirmed exactly. The training reward is unchanged, and its violation penalty also ORs over gNBs. AXIOM makes exactly the static rule's decisions in 9 of the 15 seeds, and in 5 of the other 6 it also blocks only mMTC.

S17. Fifth-review runs

This section lists every arm of the runs that answer the fifth internal review, whose plan was committed before any of them as commit c1f8cd2. Four of its six predictions failed, and every result is reported. VDN keeps DQN-I's per-gNB networks, optimiser, exploration, and target

Table 7: Env B with nineteen gNBs, 15 seeds, hex layout. SLA columns give priority-slice compliance, the mean of URLLC and eMBB compliance, per gNB and at cluster level

Policy	Reward	Per-gNB SLA	Cluster SLA	Collapsed
AXIOM	80.28	77.3%	24.7%	1
DQN-SA	74.74	66.4%	18.0%	14
DQN-I	71.20	66.8%	16.7%	0
MAPPO	59.07	71.8%	20.8%	0
Static rule	81.59	79.9%	23.0%	0
Accept all	74.77	66.4%	18.0%	15
Tuned local rule, carried over	84.09	84.5%	55.7%	0

schedule, but trains them on one joint target per step, regressing the sum of the step's chosen values on the reward plus the summed next-step maxima. Its settings were carried over unchanged, with no tuning. The per-gNB penalty charges the violation penalty of each slice once for every gNB that violates it, instead of once for the cluster, for every arm in training and evaluation. Under uneven load each gNB's load multiplier is drawn between 75% and 125% per seed, and the rule re-tuned on those training streams sheds eMBB once the URLLC queue passes 0.05 rather than 0.2.

Table 8: VDN against AXIOM and DQN-I, reward per step and priority-slice compliance, 30 seeds per cell

Cell	VDN	AXIOM	DQN-I	VDN SLA	AXIOM SLA	DQN-I SLA
Env A, 3 gNBs	12.664	12.698	11.587	99.0%	98.9%	92.9%
Env B, 3 gNBs	8.552	8.832	7.794	62.5%	65.1%	59.0%
Env A, 19 gNBs, hex	86.428	89.161	76.388	83.4%	97.0%	56.4%

Table 9: Env A with nineteen gNBs on the hex layout under higher offered load, 15 seeds. SLA is priority-slice compliance

Policy	Reward, 110%	SLA, 110%	Collapsed	Reward, 120%	SLA, 120%	Collapsed
AXIOM	84.58	63.4%	0	77.44	21.7%	0
DQN-SA	72.81	34.3%	15	65.75	17.1%	14
DQN-I	71.36	29.5%	0	68.88	15.3%	0
MAPPO	63.72	39.1%	0	61.50	19.2%	0
Static rule	85.40	65.5%	0	77.93	21.9%	0
Accept all	72.81	34.3%	15	69.25	17.6%	15

Table 10: Env B with nineteen gNBs under raised load, 15 checkpoints, neither retrained nor re-tuned. Reward per step and per-gNB priority-slice compliance

Load	AXIOM	Tuned rule	Static rule	AXIOM SLA	Tuned SLA	Static SLA
Nominal	80.28	84.09	81.59	77.3%	84.5%	79.9%
110%	77.01	70.79	78.84	41.6%	56.7%	41.6%
115%	76.59	69.04	78.50	33.4%	46.7%	33.1%
120%	76.30	67.64	78.27	26.8%	38.5%	26.4%

Table 11: Env B with seven gNBs on the ring layout under uneven per-gNB load, 15 seeds. SLA is per-gNB priority-slice compliance

Policy	Reward	SLA	Collapsed
AXIOM	20.22	63.4%	0
DQN-SA	19.04	62.3%	0
DQN-I	18.65	61.0%	0
MAPPO	15.95	59.0%	0
Static rule	20.49	64.0%	0
Accept all	18.03	56.9%	15
Tuned rule, carried over	19.86	71.6%	0
Tuned rule, re-tuned	20.17	71.8%	0

Table 12: Env B with nineteen gNBs on the hex layout with the per-gNB SLA penalty, 15 seeds. Reward is under that penalty, and SLA is per-gNB priority-slice compliance

Policy	Reward	SLA	Collapsed
AXIOM	23.48	79.2%	0
DQN-SA	-8.69	65.5%	10
DQN-I	-20.59	63.8%	0
MAPPO	-45.64	60.6%	0
Static rule	23.59	79.9%	0
Accept all	-0.28	66.4%	15
Tuned rule, carried over	39.41	84.5%	0

S18. Throughput and delay

The evaluation logs record decisions, rewards, and SLA flags but not the PRBs served or the backlog, so the usual evaluation of every stored checkpoint and rule was re-run with a recorder on the simulator's per-step output, under the plan committed as e0184c8. Every re-run whose stored log could be compared reproduced it exactly in reward, blocks, and SLA compliance, and accept-all in Env A, which has no stored log at these seeds, is marked unchecked. Throughput is the share of offered demand that is served. The delay is the mean backlog divided by the mean served PRBs, in control periods of 5 s, with the 95th percentile taken over steps and gNBs. It is a queueing proxy from the fluid model, not packet latency.

Table 13: Served share of offered demand and backlog delay in control periods, mean and 95th percentile, per slice

Policy	URLLC served	eMBB served	URLLC delay	p95	eMBB delay	p95
Env A, three gNBs						
AXIOM	99.8%	99.7%	0.11	0.66	0.11	0.47
DQN-SA	99.5%	99.4%	0.20	0.96	0.18	0.83
DQN-I	99.4%	98.5%	0.23	1.11	0.43	6.24
MAPPO	98.4%	90.5%	0.38	3.31	3.81	19.67
Static rule	99.8%	99.9%	0.10	0.59	0.05	0.31
Accept all, unchecked	97.5%	97.6%	0.80	3.01	0.76	2.65
Env A, thirty-seven gNBs, hex						
AXIOM	99.8%	99.0%	0.08	0.42	0.34	1.12
DQN-SA	94.2%	95.2%	3.61	16.87	2.09	14.28
DQN-I	97.7%	97.5%	0.76	5.23	0.72	5.49
MAPPO	96.4%	85.7%	0.58	5.38	6.28	35.70
Static rule	100.0%	100.0%	0.03	0.17	0.02	0.09
Accept all, unchecked	98.7%	98.8%	0.43	2.14	0.38	1.83
Env B, three gNBs						
AXIOM	98.4%	88.5%	0.65	2.32	3.18	19.28
DQN-SA	98.2%	82.1%	0.75	3.15	5.73	49.79
DQN-I	98.6%	82.7%	0.66	2.81	4.73	49.34
MAPPO	97.2%	83.5%	1.10	4.00	4.57	43.82
Static rule	96.0%	96.1%	1.59	5.25	1.57	5.11
Tuned rule	99.0%	84.8%	0.58	1.96	3.57	38.55
Accept all	92.8%	92.9%	2.59	7.72	2.58	7.59
Env B, nineteen gNBs, hex						
AXIOM	96.6%	92.6%	1.35	4.67	3.41	16.01
DQN-SA	93.8%	93.9%	2.26	6.99	2.24	6.82
DQN-I	94.1%	86.1%	2.17	9.71	4.17	72.77
MAPPO	89.2%	88.3%	2.66	33.84	2.58	28.13
Static rule	96.8%	96.9%	1.33	4.65	1.32	4.50
Tuned rule, carried	99.1%	87.9%	0.51	1.77	2.78	25.40
Accept all	93.8%	93.9%	2.25	6.97	2.24	6.82